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Paleomagnetism of upper Precambrian to lower Paleozoic intrusive rocks from Colorado

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ABSTRACT

The paleomagnetism of the Iron Mountain–McClure Mountain alkalic intrusive complex in the northern Wet Mountains of Colorado has been investigated. Published radiometric ages for the complex, and for similar intrusive rocks nearby, range between 485 and 704 m.y. Eighty-five samples were collected from 35 sites, and the samples were demagnetized using thermal and alternating-field techniques. In many samples thermal cleaning has yielded better isolation of the primary magnetization, but the two methods yield very similar directions. The resulting directions have a trimodal distribution. One of the three groups of direction could possibly be interpreted as a later (Ordovician or late Paleozoic?) magnetization (pole position at lat 48°N, long 107°E) and agrees well with the previous findings of Larson and Mutschler, who studied these and similar intrusive complexes. However, the other two groups of directions observed in this study are interpreted with confidence as representing late Precambrian to Cambrian magnetizations. Their pole positions are 30° apart (lat 5°N, long 174°E and lat 15°N, long 142°E) and may well reflect a trend of apparent polar wandering during this time. When averaged, these two groups have a mean direction of 104° (declination) and −2° (inclination) for 21 sites (48 samples), $k = 9$, and $\alpha_{95} = 11°$. This direction yields a pole position at lat 12°N, long 156°E ($dp = 5.5°$, $dm = 11°$), which is consistent with published Cambrian poles, but considerably different from the pole obtained by Larson and Mutschler.

INTRODUCTION

In recent years, the hypothesis by Wilson (1966) that a proto–Atlantic Ocean existed between Europe and North America during early Paleozoic time has steadily gained acceptance, and various papers have discussed the geologic evidence for this ocean (for example, McKerrow and Ziegler, 1972). Although present paleomagnetic results for early Paleozoic time are consistent with such a hypothesis, they are few in number and sometimes of questionable reliability. We report here the results of a paleomagnetic investigation of late Precambrian to Cambrian intrusive rocks in Colorado, which confirm and expand the data presented in a recent paper by Van der Voo and others (1976) on the paleomagnetism of the Upper Cambrian Wilberns Formation in Texas. These newer results are intended to contribute to a broad paleomagnetic data base necessary for a quantitative approach in the delineation of the extent, spreading history, and final consumption of the early Paleozoic ocean.

Earlier paleomagnetic studies of these and similar intrusive complexes in Colorado were published by Larson and Mutschler (1971). The present study differs from their work in that our sampling was more extensive within a geographically more restricted area. Analysis of carefully designed demagnetization experiments has enabled us to isolate three groups of directions of characteristic magnetization, two of which appear to be present in the data of Larson and Mutschler as well. A different means of averaging our (trimodal) distribution, however, leads to geomagnetic pole positions that differ from their pole, and this in turn results in a different interpretation.

GEOLOGIC SETTING

There are three alkalic intrusive complexes in the northern Wet Mountains of Colorado: the Iron Mountain–McClure Mountain complex, the Gem Park complex, and the Democratic Creek complex. All are within 50 km of each other and were emplaced during late Precambrian to Cambrian time (Brock and Singewald, 1968; Fenton and Faure, 1970; Table 1). The country rocks are Precambrian granite and metamorphic rocks of the Idaho Springs Formation.

The Iron Mountain–McClure Mountain complex has been extensively studied by Parker and Hildebrand (1963), Shawe and Parker (1967), and Heinrich and Dahlem (1969). It is the subject of ongoing investigations by Heinrich and Alexander (University of Michigan) and by the U.S. Geological Survey. The central complex is 6 to 8 km across and consists mostly of gabbro, syenite, and nepheline syenite. Paleomagnetic sampling (Fig. 1) has been concentrated on the dikes associated with the complex. These dikes occur as much as 20 km from Iron and McClure Mountains and are found mostly on the western side of the complex. The dikes range in width from 5 cm to 3 m and in length from 3 m to more than 1,000 m. More than 100 dikes have

TABLE 1. RADIOMETRIC DATES FOR THE IRON MOUNTAIN AREA

Location	Analysis	Sample	Date (m.y. B.P.)	Reference
Mount Tyndall	K-Ar	Biotite rock in stock	500 ± 25	Brock and Singewald (1968)
Mount Tyndall	K-Ar	Whole rock, phaneritic syenite dike	485 ± 24	Brock and Singewald (1968)
Mount Tyndall	K-Ar	Dike rock with devitrified syenite glass	427	Brock and Singewald (1968)
Mount Tyndall	Pb-α	Zircon, syenite stock	580 601	Jaffe and others (1959)
Mount Tyndall	Pb-α	Zircon, syenite stock	595	Christman and others (1959)
McClure Mountain	Rb-Sr	Syenite stock	517 ± 14 704 ± 70	Fenton and Faure (1970)

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been mapped by Dahlem (1965), and Alexander has mapped over 100 more. The sequence of intrusion of the complex is, from oldest to youngest, (1) peridotite, (2) gabbro and associated dikes, (3) hornblende syenite and associated dikes, (4) ijolite, (5) nepheline syenite, (6) lamprophyre dikes, and (7) carbonatite (Heinrich and Dahlem, 1969).

The Ilse fault is 10 km to the east of Iron and McClure Mountains, and the Texas Creek fault is 15 km to the west (Fig. 1). These are the only major faults that have been mapped in the area. There is no petrologic evidence of a thermal event since the intrusion of the dikes, which would have introduced a partial thermoremanent magnetization, nor is there evidence of any structural disturbance, though this can never be ruled out completely.

Samples were taken at 35 sites throughout the complex. Twenty-eight of these were from outlying dikes of the Iron Mountain–McClure Mountain complex, and four were from gabbros within the complex. The other three sites were in the Gem Park complex, two of which were in the main body of gabbro there (Fig. 1).

Sixty-two samples from 12 sites (prefixed FCA in Table 2) were drilled with a portable drilling apparatus by French and Van der Voo in August 1973. They were all from dikes about 10 km west of the center of the Iron Mountain–McClure Mountain complex. During the summer of 1975 a collection of more than 300 samples was made by Alexander for petrologic studies. Twenty-three oriented hand samples (prefixed ACC in Table 2) from 23 sites formed part of this second collection. All samples were oriented with a Brunton compass, and a solar compass was used to check the orientations (for details, see French, 1976). Thus, the total collection represents 29 dikes and 6 sites of the gabbro complexes,

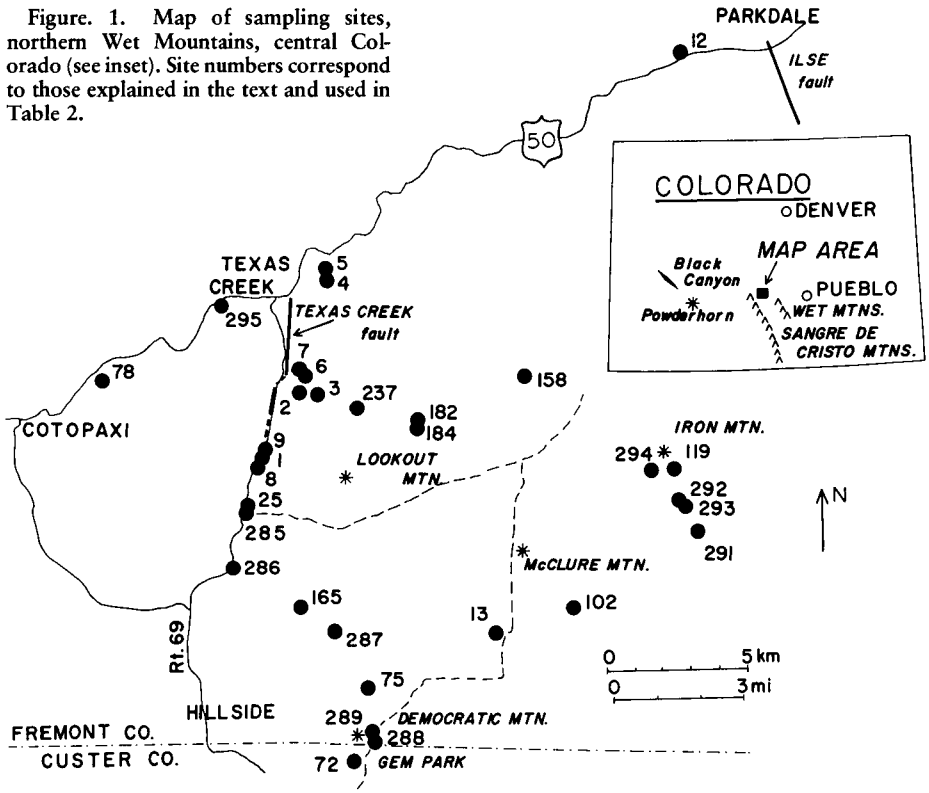


Figure. 1. Map of sampling sites, northern Wet Mountains, central Colorado (see inset). Site numbers correspond to those explained in the text and used in Table 2.

distributed over some 30 km², with a majority from the western side of the complexes.

Petrologic identifications of the dikes include various gabbro, teschenite, syenogabbro, trachyte, and peridotite. The principal carrier of magnetization is magnetite, which composes as much as 10% of some samples. In microscopic studies most of the magnetite appeared to be primary, which is further corroborated by electron microprobe studies that indicated significant amounts of Ti, Al, Mg, and Cr present in the magnetites. In most of the samples the plagioclase is somewhat altered. Considerable amounts of secondary chlorite, epidote, and calcite are often present. Sometimes, the magnetite is slightly oxidized, so that some secondary hematite is occasionally observed, but in microscopic studies primary hematite appeared to be present as well, mostly in the feldspars. However, the blocking temperatures and coercivities observed during demagnetization indicate that hematite carries little of the magnetization and that the principal magnetic mineral is magnetite.

We have assumed a late Precambrian to

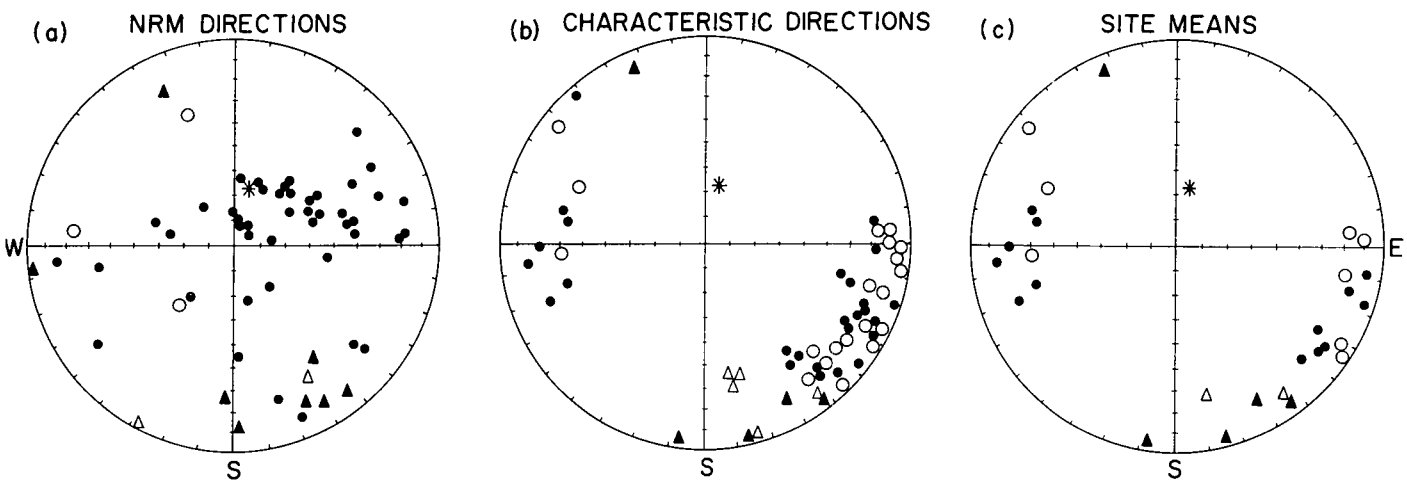


Figure 2. Equal-area projections of the directions of magnetization from the Colorado intrusive rocks. Open symbols pointing upward, solid symbols pointing downward. Asterisk is local present-day geomagnetic field direction. Dots and circles: groups I and II; triangles: group III. Each symbol in (a) and (b) represents the direction obtained from the corresponding sample used in the computation of the site-mean directions (c).

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Cambrian age for these dikes. This is based primarily on a whole-rock Rb-Sr date from the McClure Mountain stock (Fenton and Faure, 1970). This date, taken from an isochron using five samples, gives a minimum age of 517 m.y. Discarding one sample, an age of 704 ± 25 m.y. is obtained.

There are several dates from the neighboring complexes (summarized in Table 1). Mount Tyndall is in an alkalic intrusive complex located 20 km to the southeast of the McClure Mountain stock and has been dated by Brock and Singewald (1968). Their K-Ar dates include a biotite age for the Mount Tyndall stock of 500 ± 25 m.y. and a whole-rock age for a phaneritic syenite dike of 485 ± 24 m.y. There are also lead-alpha ages on syenitic rocks within this stock of 580, 595, and 601 m.y. (Christman and others, 1959; Jaffe and others, 1959).

In view of the variety in directions of magnetization observed in this study, our collection of samples may well span the entire period between 704 and 485 m.y. B.P. This implies mostly late Hadrynian (700 to 600 m.y. B.P.) and Cambrian (600 to 500 m.y. B.P.) ages, although some sites may be as young as Early Ordovician.

TABLE 2. SUMMARY OF SITE MEAN DIRECTIONS OF CHARACTERISTIC MAGNETIZATION

Site*		N†	Declination (°)	Inclination (°)	k#	α_{95}# (°)	Remarks**
Sites with characteristic directions of group I							
FCA	1A	6	124	−1	33	11.8	AF
FCA	2A	7	121	−7	9	21.7	AF
FCA	2B	1	121	+21	..	..	AF
FCA	3A	3	126	+14	7	53.0	AF and thermal
FCA	3B	3	105	+14	22	26.7	Thermal
FCA	5	3	132	+19	21	27.7	AF and thermal
FCA	6	5	124	+13	26	15.3	AF and thermal
FCA	7	5	99	+7	11	24.4	Thermal
ACC	75	1	308	−8	..	..	Thermal
ACC	102	1	108	+4	..	..	AF
ACC	287	1	294	−32	..	..	Thermal
ACC	294	1	266	−30	..	..	AF and thermal
Sites with characteristic directions of group II							
FCA	4	3	100	−18	8	46.0	AF and thermal
ACC	13	1	86	−17	..	..	AF and thermal
ACC	72	1	269	+20	..	..	AF and thermal
ACC	119	1	89	−10	..	..	Thermal
ACC	158	1	254	+30	..	..	AF and thermal
ACC	184	1	279	+32	..	..	Thermal
ACC	291	1	250	+20	..	..	Thermal
ACC	292	1	264	+13	..	..	AF
ACC	293	1	283	+29	..	..	AF
Sites with characteristic directions of group III							
FCA	1B	4	168	−26	29	17.3	AF and thermal
ACC	25	1	143	+6	..	..	AF and thermal
ACC	78	1	152	+16	..	..	Thermal
ACC	285	1	337	+8	..	..	Thermal
ACC	286	1	142	−9	..	..	AF and thermal
ACC	288	1	188	+4	..	..	AF and thermal
ACC	289	1	165	+3	..	..	AF and thermal
Mean direction for group I		12	116	+13	22	10.0	Pole, lat 15°N, long 142°E
Mean direction for group II		9	88	−21	39	8.0	Pole, lat 5°N, long 174°E
Mean direction for group III		7	159	−2	15	16.0	Pole, lat 48°N, long 107°E
Mean direction for groups I and II		21	104	−2	9	11.0	Pole, lat 12°N, long 156°E

Note: Sites with directions conforming to the present-day geomagnetic field directions (remagnetized): FCA 8, 9; ACC 12, 165, 182, 237, 295 (10 samples); both AF and thermal.

* Site numbers correspond to Figure 1.

† *N* is the number of samples used for the computation of the mean direction of the corresponding site (10 samples were not used because they showed only present-day field directions; 17 samples were not used because no characteristic direction of magnetization could be accurately determined).

k and α_{95} are the parameters of the statistical treatment (Fisher, 1953), given only where the number of samples exceeded three.

PALEOMAGNETIC RESULTS

Natural remanent magnetization (NRM) directions were widely scattered but showed a concentration in the northeast quadrant of the lower hemisphere. These directions are plotted in Figure 2a. NRM intensities ranged between 2×10^{-4} and 10×10^{-4} emu/cm^3, though a few samples had intensities as high as 10^{-1} emu/cm^3. All samples were progressively demagnetized using alternating-field (AF) demagnetization, thermal demagnetization, or both. All samples had at least two components of magnetization, and these were analyzed using the Zijderveld (1967) method. Directions of the hard components (with high blocking temperatures or high coercivity) and soft components were both calculated. In view of the challenging complexity of the observed directions of magnetization, the importance of establishing the reliability of these directions, and the difference between our interpretation and that of Larson and Mutschler (1971), the demagnetization will be discussed and illustrated in a rather elaborate manner.

Thermal Demagnetization

Fifty-two specimens (44 samples) from 28 sites were thermally demagnetized. Demagnetization characteristics and blocking temperatures varied widely so that all specimens were treated with at least six steps. Four demagnetization diagrams displaying different behavior are shown in Figures 3 and 4.

In specimen FCA 56A (Fig. 3a) a small component of magnetization was removed between 20 and 400°C. This component had a declination *(D)* of 350° and an inclination *(I)* of 82° (close to the present-day geomagnetic field in Colorado) and an intensity of 18% of the total NRM; this magnetization is assumed to be secondary and of recent origin. A magnetization with a direction of $D = 91°, I = -5°$ is removed between 500 and 575 °C. This direction was chosen since little or no secondary magnetization appeared to remain in this trajectory, and the decay was approximately linear toward the origin.

Sample 51A (Fig. 3b) also had a secondary magnetization, with a direction close to

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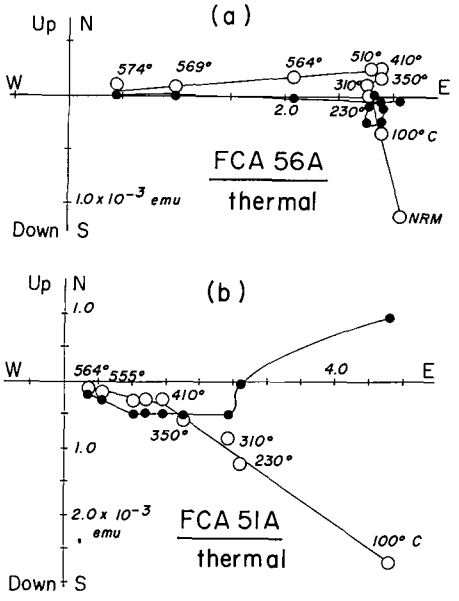


Figure 3. Orthogonal projections (demagnetization vector diagrams) of the successive end points of the magnetization vector during progressive thermal demagnetization. Solid symbols represent projections on the horizontal plane, open symbols on the east-west vertical plane. Samples from sites 7 and 6 (group I).

that of the present-day field, which was removed fully by 500 °C. In this sample, the intensity of the soft component, 89% of the total NRM intensity, was much larger than that of the characteristic component. Because the harder magnetization appears to decay linearly to the origin, we assume the remaining characteristic component (D = 115°, I = +19°) to be free from contamination by secondary magnetizations. Thermal demagnetization plots for ACC 286B and ACC 285B are shown in Figure 4. These samples, taken from dikes slightly more than 2 km apart, have very similar decay characteristics, but with mutually reversed directions. Both samples have relatively strong characteristic magnetizations, and their secondary magnetizations were easily removed below 300 °C.

Alternating-Field Demagnetizations

Ninety-five specimens (77 samples) from 33 sites were demagnetized in alternating fields of as much as 1,000 Oe. The AF results agree well with the thermal results, but in some cases the samples were not cleaned by 1,000 Oe. Moreover, care had to be taken to minimize the effects of spurious magnetizations introduced in the demagnetization process above 500 Oe (French, 1976). Demagnetization characteristics again varied considerably from site to site. Most samples lost 50% of their intensity by 100 Oe and 90% by 500 Oe. Specimen FCA 50B (Fig. 5) is one of the better (less unstable) specimens having this typical behavior. From this specimen, two secondary components of magnetization were removed, one below 15 Oe, the other between 25 and 100 Oe. The harder, characteristic component was isolated above 130 Oe. The approximately straight decay to the origin of the magnetization vector above 170 Oe indicates that, in all likelihood, there is only one component of magnetization being removed. The hard component of this specimen has a characteristic direction of D = 119°, I = +7° (the direction of the magnetization remaining in the sample at 170 Oe). The soft magnetization has a direction of D = 59°, I = +49°, which is intermediate between the present-day field direction (D = 14°, I = +67°) and the hard direction. This specimen is from the same site as FCA 51 (D = 115°, I = +19°), and the AF and thermal demagnetization techniques yield confirming results, as can be seen by comparing the final directions above 550 °C (Fig. 3b) and 170 Oe (Fig. 5, FCA 50B). Roughly half the AF demagnetizations followed a similar pattern with similar coercivities.

Another group of samples had components of magnetization with higher coercivities. Specimen FCA 31B (Fig. 5) is typical of these. A very soft component was removed below 50 Oe, but the direction continued to move with fields of as much as 600 to 800 Oe, in which an intermediate component persisted. By 800 to 1,000 Oe, apparently only the hard component remained, and the magnetization appeared to decay to the origin.

Figures 6 and 7 compare thermal demagnetizations with AF demagnetization from the same samples. Figure 6 shows sample ACC 72 with low blocking temperatures and relatively high coercivities, and Figure 7 shows sample ACC 158 with higher blocking temperatures and low coercivities. Thermal and AF demagnetizations reveal the same secondary and characteristic directions for each sample.

Many samples were not cleaned by 1,000 Oe and did not have a clear decay to the origin. Such a sample, specimen FCA 53B, is shown in Figure 8b, along with the thermal demagnetization from the same sample (FCA 53A) plotted in Figure 8a. Because of this incomplete cleaning with alternating fields, thermal results were generally preferred. In cases where only AF demagnetization results of such unsatisfactory nature were available, they have not been used for computation of the mean direction (17 out of the 85 samples). Also, some of the samples acquired spurious magnetizations in the demagnetization process. Care was taken to minimize this effect; these procedures are described in French (1976).

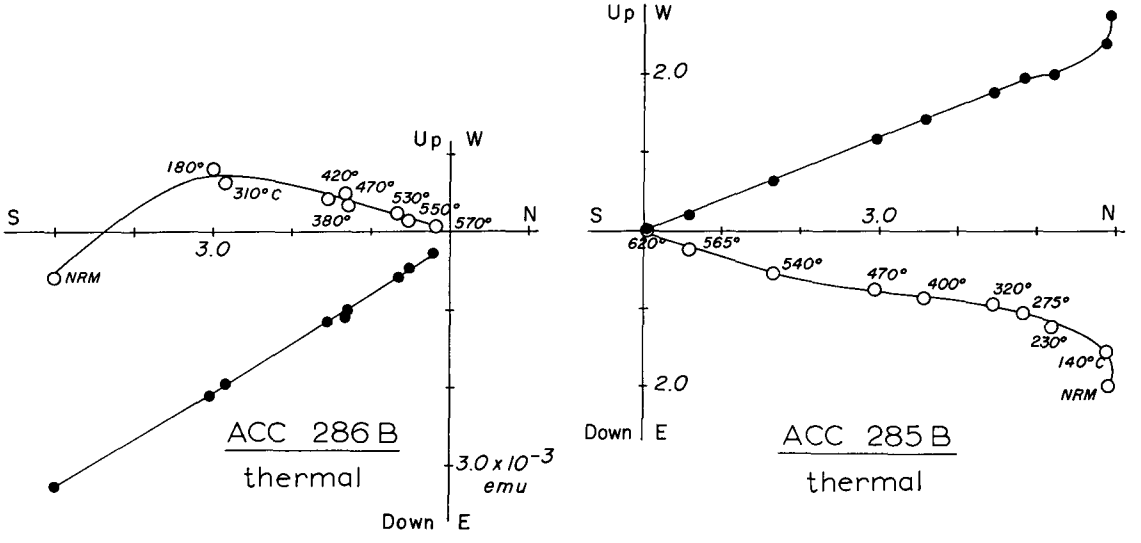


Figure 4. Thermal demagnetization vector diagrams of samples from sites 286 and 285. For explanation of symbols see Figure 3, although vertical projections here are on the north-south vertical plane in order to illustrate the nature of these group III directions more clearly.

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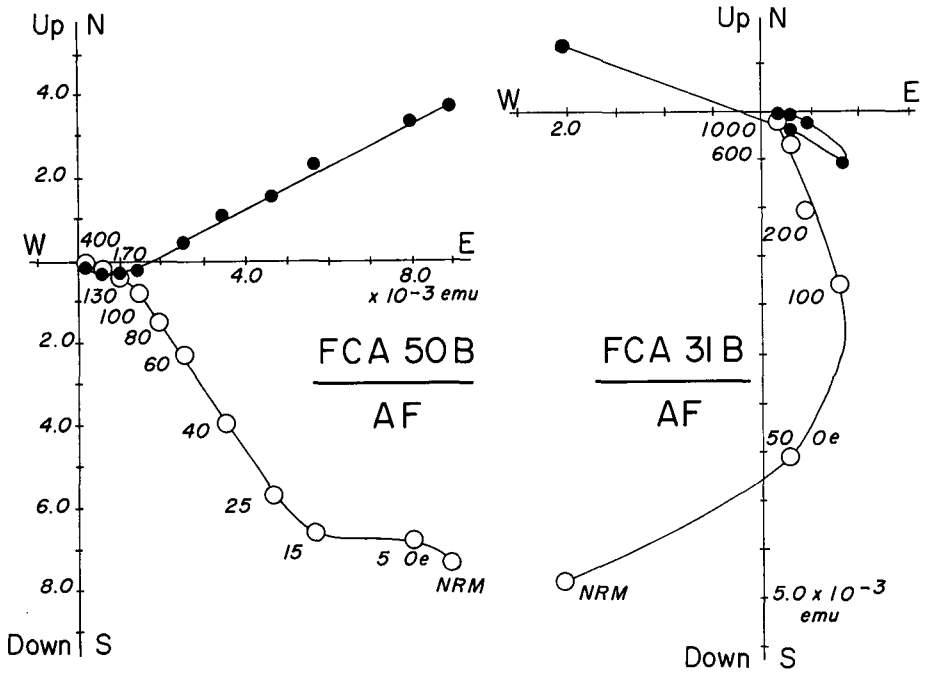


Figure 5. AF demagnetization vector diagrams of samples from sites 6 and 3A (group I). For explanation of symbols see Figure 3.

Resulting Directions

The characteristic directions of the samples are plotted in Figure 2b, together with the site mean directions (Table 2, Fig. 2c). The site means are replotted in Figure 9 together with a density contour plot of those means. The plots are equal-area projections of the southeast hemisphere (see caption). The density plot shows at least two distinct groups of directions, and we have (maybe arbitrarily) divided the directions into three groups. Group I has a mean direction of $D = 116°, I = +13°$ ($N = 12, k = 22, \alpha_{95} = 10°$, pole position at lat 15°N, long 142°E). Group II has a mean direction of $D = 88°$, $I = -21°$ ($N = 9, k = 39, \alpha_{95} = 8°$, and pole position at lat 5°N, long 174°E). Group III has a mean direction of $D = 159°, I = -2°$ ($N = 7, k = 15, \alpha_{95} = 16°$, pole position at lat 48°N, long 107°E) and forms a broad streak away from the group I directions. Seven sites (10 samples) had directions that did not fall in these groups, and they conformed mostly to the present-day geomagnetic field direction; these samples are assumed to be remagnetized in recent times and have not been utilized.

There appears to be little petrologic correlation between the rock types and the groups of directions mentioned above. The best correlation of directions with petrology possibly exists for groups I and II, with group I tending to have more of the nepheline syenite, syenite, and lamprophyre dikes, whereas group II includes mainly gabbroic rocks. No good correlation exists for group III.

Group III directions may not represent the late Precambrian to Cambrian geomagnetic field. Samples at four of the seven sites in this group were taken from within 100 m of the Texas Creek fault and show some

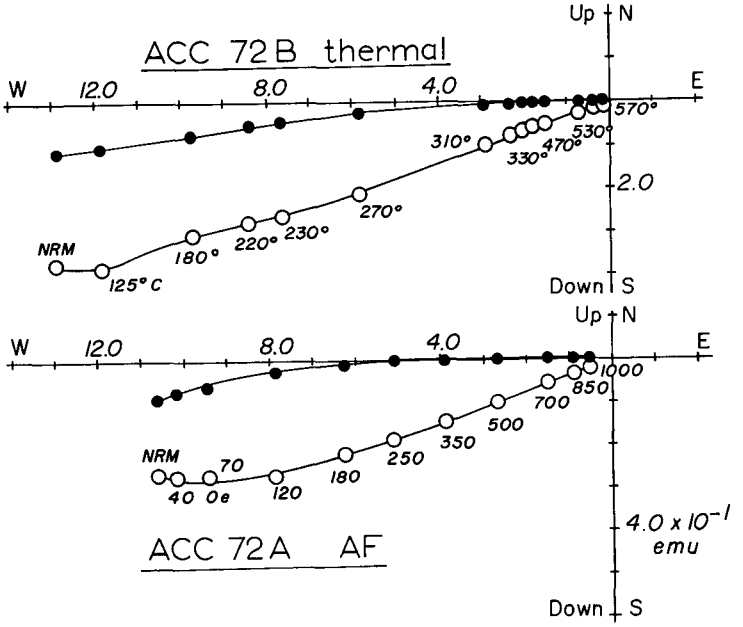


Figure 6. Thermal and AF demagnetization vector diagrams of specimens from the same sample of site 72 (group II). For explanation of symbols see Figure 3.

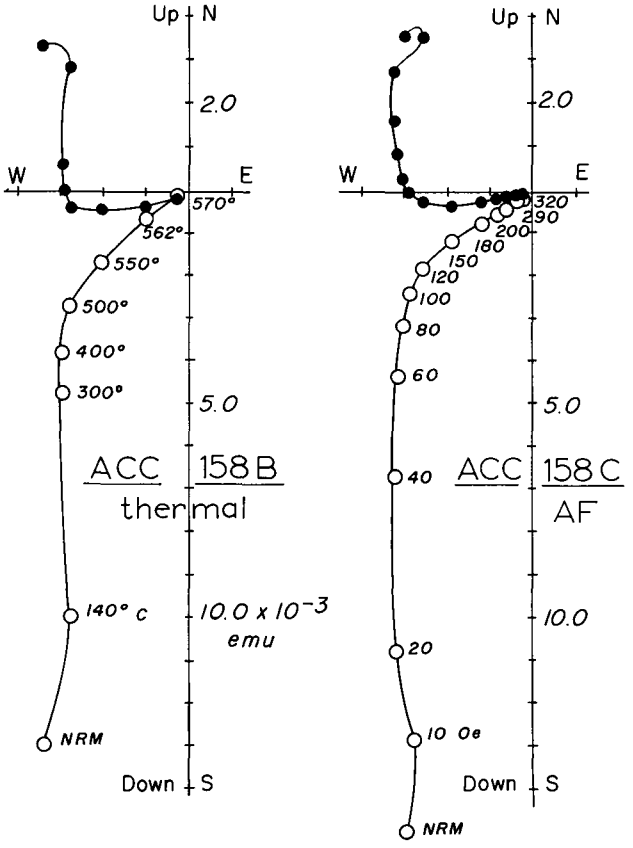


Figure 7. Thermal and AF demagnetization vector diagrams of specimens from the same sample of site 158 (group II). For explanation of symbols see Figure 3.

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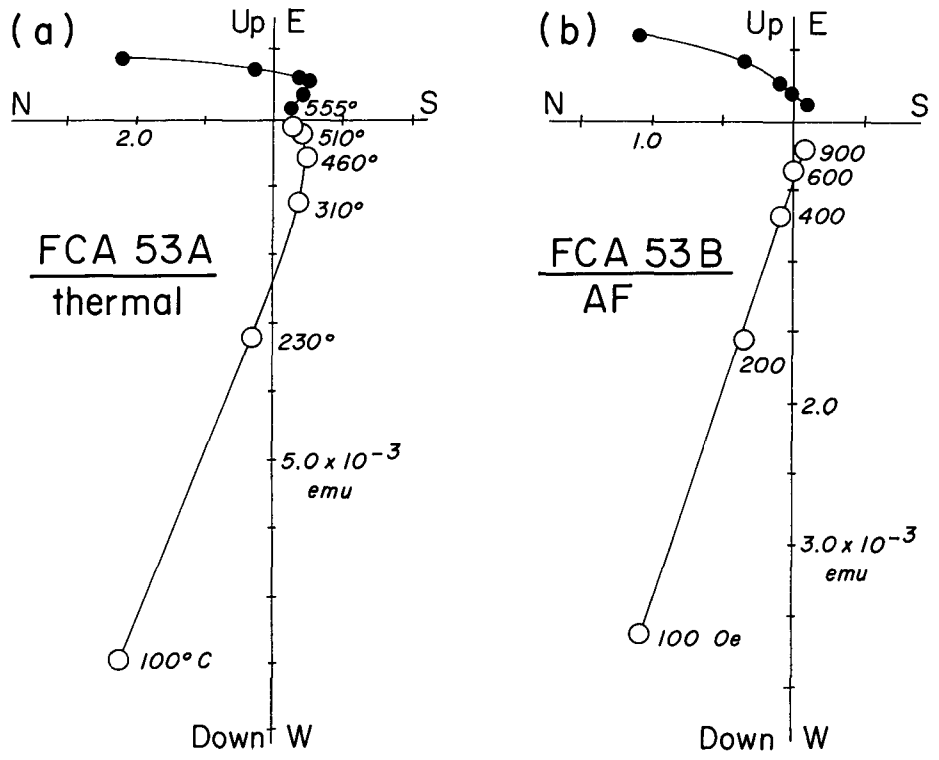


Figure 8. Thermal and AF demagnetization vector diagrams of two specimens from the same sample of site 6 (group I). For explanation of symbols see Figure 3, although vertical projections here are on the north-south vertical plane to illustrate the nature of the secondary magnetizations (conforming to the present-day field direction).

alteration, but no more so than several dikes with the directions of groups I and II. Two other sites in group III were in fresh gabbro of the Gem Park complex. The resulting pole position for group III is within 5° of the Late Permian pole at lat 49°N, long 113°E (Van der Voo and French, 1974) and does not fall near other Cambrian or older poles. If these rocks were remagnetized, their original magnetization has been completely destroyed, since demagnetization never revealed a direction falling in group I or II. Had such a magnetization direction been found, superimposed on group III directions, it would be strong evidence for remagnetization.

Other explanations for this group III direction are that the rocks are younger (Early Ordovician) or that there was a loop in the polar-wander path during late Precambrian or Cambrian time. Group III sites could then be older or younger than groups I and II. This explanation could also account for the difference in directions between groups I and II. Further consideration of this hypothesis will be deferred to the discussion.

Groups I and II are taken to be primary late Precambrian to Cambrian thermoremanent magnetizations. This is based on four factors. First, these are the components of magnetization with the highest blocking temperatures and the highest coercivities. Second, reversed and normal directions are present in each group. Third, if the directions of groups I and II were due to a later remagnetization, we would expect the pole positions to lie somewhere on the polar-wander path for a later time. The pole positions from groups I and II are inconsistent with any poles younger than Late Cambrian, as is discussed below. Finally, there has been no apparent widespread thermal event in the area since the intrusion of the complex that could have completely remagnetized the rocks.

The occurrence of the two distinct groups of directions (I and II) is difficult to explain with any certainty. The pole positions for the two groups fall 30° apart. The difference could however be explained by different times of intrusion, since the available radiometric dates show a rather large spread. At present, it seems best to interpret both groups as late Precambrian to Cambrian magnetizations, with resulting pole positions at lat 5°N, long 174°E and lat 15°N, long 142°E. If the sites of these groups are averaged together, their mean direction, with $D = 104°$, $I = -2°$ ($N = 21$, $k = 9$, $\alpha_{95} = 11°$), yields a pole position at lat 12°N, long 156°E.

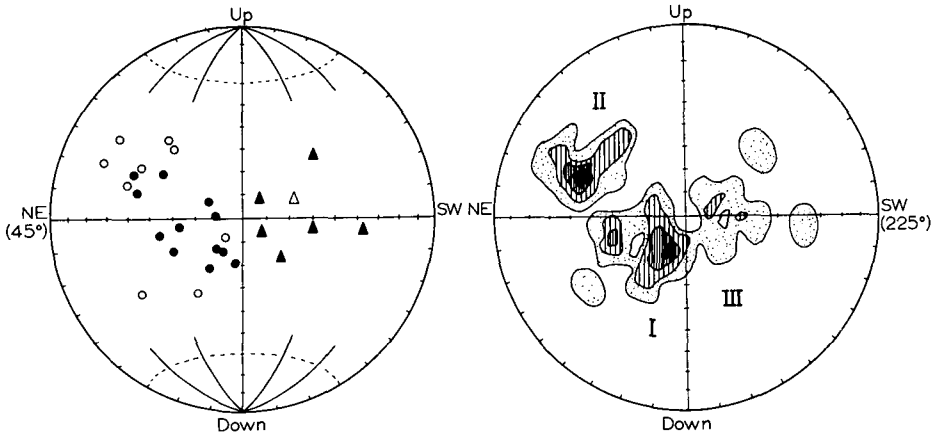


Figure 9. Site mean directions and density plot of these directions. Hemispherical equal-area projection. Note the unusual projection plane with a vertical, northeast-southwest orientation. This projection plane has been chosen to illustrate the site mean directions in a more continuous manner. All symbols indicate directions in the southeastern hemisphere, but open symbols represent directions that have been reversed before projection. Contour intervals are 0%, 3.5%, 7%, and 10.5% of the data set (28 entries). The density contouring method involved a counting area of 1% of a hemisphere (circular on the hemisphere and projected circle on the equal-area plot).

PREVIOUS STUDIES

Mutschler and Larson (1969) and Larson and Mutschler (1971) studied the paleomagnetism of some of the same intrusive complexes discussed in this study. They also collected 13 samples from two similar alkalic intrusive complexes 150 km to the west. These other samples were from syenite and diabase dikes in the Black Canyon area and from a gabbroic dike in the Powderhorn district (Fig. 1, inset). Radiometric dates suggest similar ages (525 to 583 m.y.; Jaffe and others, 1959) for the Powderhorn complex.

Larson and Mutschler obtained results similar to ours, although their interpretation is somewhat different. They used both AF and thermal demagnetization with more or less satisfactory results. With AF cleaning they found that some of their samples cleaned nicely below 400 Oe, whereas others had strong secondary magnetizations remaining. Their AF direction is $D = 143°$,

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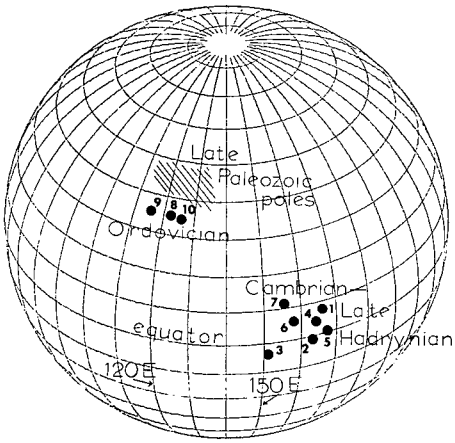


Figure 10. Late Hadrynian, Cambrian, and Ordovician pole positions for North America (from Van der Voo and others, 1976). Numbers correspond to those in Table 3.

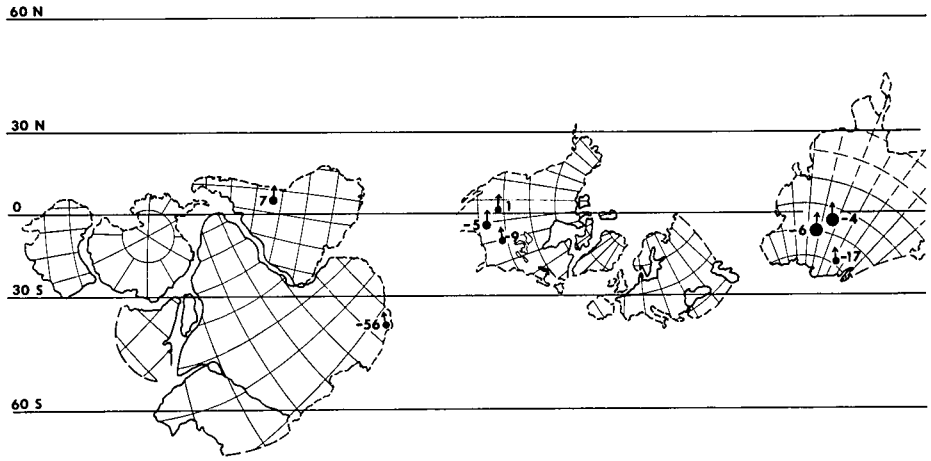


Figure 11. Paleogeographic map in Mercatur projection for Late Cambrian time. The major continents are plotted according to their observed (or interpolated) paleomagnetic results (after French, 1976). Arrows represent declinations, which in an ideally consistent situation would all be perpendicular to the latitude lines. The latitude values, as indicated near the collecting localities, were calculated from the paleomagnetic inclinations. A degree of freedom exists in this projection, in that relative longitudes of the continents are indeterminable by paleomagnetism; thus, continents can be shifted eastward or westward, parallel to lines of latitude, without violating the paleomagnetic data.

$I = +6°$ ($N = 18$, pole at lat 36.6°N, long 122.3°E). The samples that cleaned well in AF demagnetization also cleaned well thermally. Likewise, those samples that behaved poorly during AF demagnetization remained erratic with thermal demagnetization. With temperatures of as much as 500 °C, they obtained a direction $D = 162°$, $I = +9°$ ($N = 14$, pole at lat 44.2°N, long 99.8°E). These directions agree with those of our group III. Their samples from the Wet Mountains all have a more easterly direction than those from the other sites. Owing to a different sampling bias, Larson and Mutschler (1971) did not have a majority of samples with these easterly directions, and they grouped all directions together. On the basis of our expanded collection, which shows a trimodal distribution, we feel it is better to treat the pole positions for each group as semiseparate entities.

DISCUSSION

The combined mean direction of groups I and II yields a pole at lat 12°N, long 156°E, which is fairly close to Late Cambrian poles (Van der Voo and others, 1976) listed in Table 3 and plotted in Figure 10. Late Hadrynian poles (700 to 600 m.y. B.P.) plot nearby and are statistically indistinguishable.

A peculiar observation in Cambrian rocks, however, is that directions are often found that correspond to poles near the cluster of late Paleozoic poles. These directions were observed in this study (group III), by Larson and Mutschler (1971), by Vincenz and others (1975) in the Wichita granites, by Al-Khafaji and Vincenz (1971) in the Lamotte Formation, by D. P. Elston (1976), written commun.) in the Muav Limestone, and by French (1976) in the Middle Cambrian Rome Formation of the central Appalachians. In some cases the most logical explanation seems to be a partial or complete remagnetization of these rocks during late Paleozoic time (Vincenz and others, 1975). On the other hand, the observed occurrence of both "late Paleozoic" and unambiguous Cambrian directions in a single formation or complex (the Colorado intrusive rocks, the Lamotte Formation, the Wichita granites, the Rome Formation) or in superimposed formations (the Muav Limestone overlying the Tapeats Sandstone in Arizona) renders the remagnetization hypothesis unlikely. These directions could then indicate a certain amount of apparent polar wander during or shortly after Cambrian time.

A definite choice between the two possibilities (remagnetization or apparent polar wander) appears to be premature. Detailed studies of well-dated multiple intrusion complexes or stratigraphically well-known sedimentary sections are necessary to resolve this problem.

The consequences of the ultimate choice have important implications for the opening stages of the early Paleozoic ocean thought to have existed between Europe and North America (Wilson, 1966). Figure 11 is a paleogeographic map (from French, 1976) of Late Cambrian positions of Europe, North America, Siberia, and

TABLE 3. PALEOMAGNETIC POLE POSITIONS

Pole no.	Rock group	Age	Reference	Pole position (lat) (long)
10	Beemerville alkaline complex	Late Ordovician	Proko and Hargraves (1973)	35.0°N, 126.0°E
9	Trenton Limestone	Middle Ordovician	McElhinny and Opdyke (1973)	36.0°N, 116.0°E
8	Colorado intrusive rocks	Cambrian-(Ordovician?)	Larson and Mutschler (1971)	36.6°N, 122.3°E
7	Colorado intrusive rocks	Late Precambrian to Cambrian	This study	12.0°N, 156.0°E
6	Wilberns Formation	Late Cambrian	Van der Voo and others (1976)	6.4°N, 158.8°E
5	Lamotte Formation	Late Cambrian*	Al-Khafaji and Vincenz (1971)	0.5°N, 168.0°E
4	Wichita granites	Late Cambrian*	Spall (1970)	4.0°N, 164.0°E
3	Grenville dikes	Hadrynian	Murthy (1971)	3.0°S, 151.0°E
2	Coronation sills	Hadrynian (647 m.y.)	Robertson and Baragar (1972)	1.0°S, 163.0°E
1	Franklin diabases	Hadrynian (675 m.y.)	Fahrig and others (1971)	8.0°N, 167.0°E

* Pole obtained through thermal demagnetization.

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Gondwanaland, plotted according to their observed (or interpolated) paleomagnetic pole positions. The only Cambrian poles used to position North America are those that cluster near lat 5°N, long 160°E. If Middle Cambrian pole positions are established as reliable, while falling near the late Paleozoic poles of Figure 10, a loop in the apparent-polar-wander path must be assumed, tracing from the late Hadrynian poles through the poles near the late Paleozoic cluster and back to the Late Cambrian (middle Croixian) pole of the Wilberns Formation at lat 6°N, long 159°E. Such rapid apparent polar wander will significantly alter the picture of Figure 11, essentially through a rotation of North America while it remains in equatorial positions. It must be remarked, however, that Cambrian poles from Europe are not numerous so far, and that they are rather scattered. Thus we must defer a detailed discussion of the relative drift between North America and Europe, and of the spreading history of the early Paleozoic ocean in between, until more results are available.

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